Relationship between Sleep Quality and Readiness to Change Weight-Related Health Behaviors in Smokers Unmotivated to Quit

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Background

- Smoking cessation often leads to weight gain, which may demotivate smokers from quitting[1,2]
- Improved sleep can ameliorate weight gain and nicotine withdrawal symptoms[3,4]

Study Objectives

- To determine which weight-related health behaviors smokers want to change
- To determine if the Behavioral Risk Factor Surveillance System (**BRFSS**) Inadequate Sleep Questionnaire is associated with readiness to change weight-related health behaviors

Study Design

- Data are from a randomized controlled trial that tested a novel smartphone app designed to increase motivation for smoking cessation in a sample of adults not ready to quit smoking
- Secondary analysis of baseline data
- ***N* = 152**

Table 1: Demographic Characteristics of Smokers

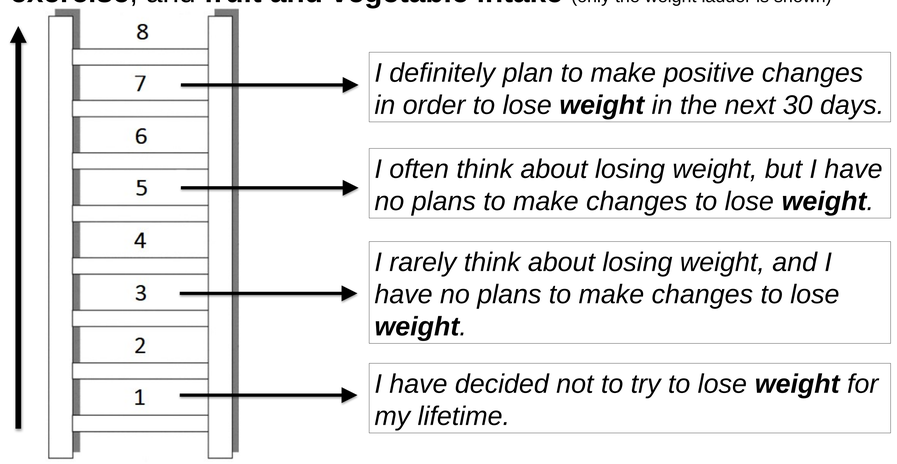
Variable	*N* (%)	Variable	M (SD) or *N* (%)
Female	103 (67.8%)	Age	50.0 (12.5)
Race		BMI (Category)*	
White	113 (74.3%)	Underweight	1 (1.8%)
Black	13 (8.6%)	Normal	15 (26.8%)
American Indian	20 (13.1%)	Overweight	10 (17.9%)
Other	6 (4.0%)	Obese	30 (53.6%)
Employed	72 (47.4%)	Weight (lbs.)*	202.1 (54)
Smokes within 5 mins of waking	67 (44.1%)	Cigarettes per day	18.1 (8.5)

*Lacking weight and BMI data for 93 participants after halting in-person visits due to the Covid-19 pandemic

This work was supported by the Oklahoma Tobacco Settlement Endowment Trust (TSET) grant 092-016-0002, and used the Mobile Health Technology Shared Resource which receives funding from the NCI Support Grant P30CA225520 awarded to the Stephenson Cancer Center.

Measures

- Readiness to Change Health Behaviors Ladders for **weight**, **exercise**, and **fruit and vegetable intake** (only the weight ladder is shown)



- Self-Rated Health question "*Would you say your health, in general, is*" (1 = "Excellent", 5 = "Poor")[5]
- 4 items from the BRFSS Inadequate Sleep Questionnaire (see Table 2) that assesses sleep duration and problems related to sleep

Table 2: Inadequate Sleep Items

BRFSS 1	BRFSS 2	BRFSS 3	BRFSS 4
During the past 30 days, for how many days have you felt you did not get enough rest or sleep?	On average, how many hours of sleep do you get in a 24-hour period?	Do you snore?	During the past 30 days, have you ever nodded off or fallen asleep, even just for a brief moment while driving?

Results

Weight-Related Behaviors and Readiness to Change Them

- See Figure 1

Sleep-Related

- Most (61%) participants endorsed snoring
- Average sleep duration was 6.7 hours (SD=1.5) in 24-hour period

Results

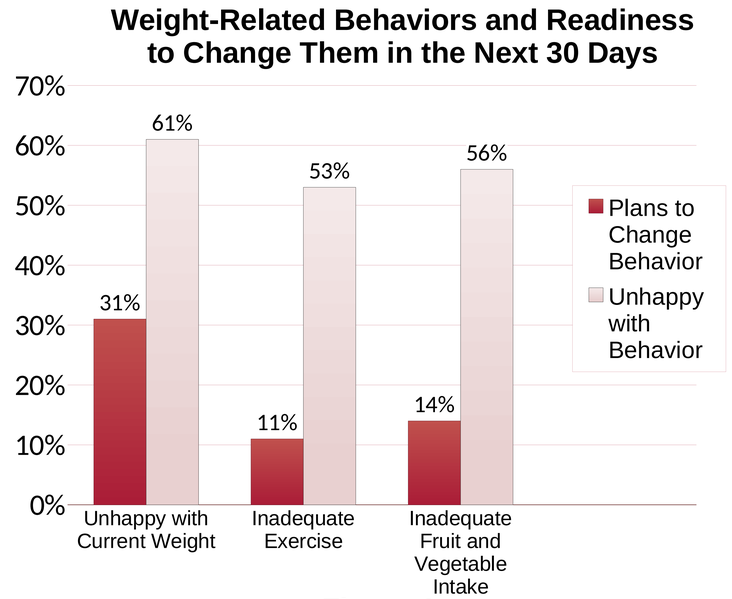


Figure 1

Conclusions

- Most smokers were dissatisfied with weight and weight-related behaviors, however, only a minority had plans to lose weight and fewer had plans to increase exercise and increase fruit and vegetable intake in the next month
- Items from the BRFSS sleep questionnaire were associated with dissatisfaction with weight and readiness to change weight-related behavior

Limitations and Future Direction

- Cross-sectional data were used; longitudinal data may reveal additional associations that may be addressed in future interventions
- Future studies should examine if sleep interventions can impact readiness to change weight-related health behaviors

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